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SUMMARY

I work on high-assurance systems security: Trusted Execution Environments, cryptographic attestation and root-of-trust, and hypervisor/OS isolation, paired with the adversarial side (side-channel attacks, exploit development, and fuzzing) to test whether those guarantees hold. Throughout my career, I have tackled security problems across the full hardware-software stack, driven by a desire to turn complex research problems into production-ready solutions.

WORK EXPERIENCE

Senior Security Engineer: NVIDIA Corporation, Santa Clara, CA **Jul 2024 – Current**

- Owned an end-to-end cryptographic attestation chain from silicon root of trust to signed evidence for Vera and Grace CC.
  - Developed cross-layer firmware and software stacks across bootloader, RMM, ATF, PSCFW, and Caliptra DPE to enable end-to-end attestation and evidence generation for Vera CCA and Baremetal Secure AI on Vera and Grace CPUs.
  - Developed and deployed CoRIM/CoMID manifests for Vera CCA and Vera/Grace baremetal attestation.
- Led Live Firmware Activation (LFA) support for RMM, building the software stack across RMM, ATF, and Vera IROT.
- Hardened guest OS boot sequence and model loading using initramfs disk encryption with early attested key provisioning.

Postdoc Researcher: Georgia Institute of Technology, Atlanta, GA **Jan 2022 – Jul 2024**

- Developed an automated fuzzer selection tool that dynamically selected fuzzers based on runtime analysis [3]
- Designed and implemented a hardware-enforced secure I/O path between confidential VMs and DMA peripherals [1]
- Conducted vulnerability research on Intel and ARM's latest hardware-isolated confidential VM architectures (TDX & CCA)

Postdoc Researcher: Korea Advanced Institute of Science and Technology, Korea **Mar 2021 – Dec 2021**

- Demonstrated an eviction-free cache side-channel attack on x86 and Apple M1, bypassing eviction-monitoring defenses [8]
- Assisted in supervising and mentoring Ph.D. students across a large research group (20+ members).

Research Intern: Georgia Institute of Technology, Atlanta, GA **Nov 2019 – May 2020**

- Developed a hardware-based side-channel mitigation system on GEM5, monitoring cache and TLB eviction events [2]

Research Intern: Microsoft Research, Redmond, WA **May 2016 – Aug 2016**

- Executed practical ROP attacks on Intel SGX by exploiting memory corruption vulnerabilities without binary details [4]
- Developed a secure CDN enabling enterprises to offload sensitive user data to untrusted third-party cloud providers.

PUBLICATION & PATENT **(830 citations)**

Top-tier conferences (SP, USENIX Security, NDSS) and Journal (TDSC)

[1] **PORTAL: Fast and Secure Device Access with Arm CCA for Modern Arm Mobile SoC** | Paper **SP 2025**

- Eliminated per-transfer encryption overhead for Arm CCA Realm VMs accessing DMA-capable peripherals.
- Extended the Linux SMMU driver to delegate page table management to RMM; implemented the Realm-side driver and the full RMI call path across KVM, TF-RMM, and the SMMU hardware controller. Modified the RMM implementation to enable hardware-enforced (GPT) isolated DMA regions without cryptographic operation.

[2] **SENSE: Enhancing Microarchitectural Awareness via Subscription-Based Notification** | Paper **NDSS 2024**

- Addressed the lack of microarchitectural visibility in TEEs, which left them blind to cache/TLB side-channel attacks.
- Extended the GEM5 out-of-order CPU pipeline with new micro-ops and added event-tracking metadata bits to the cache and TLB models; implemented the enclave-side handler dispatch path.

[3] **autofz: Automated Fuzzer Composition at Runtime** | Paper | Hacker News **USENIX 2023**

- Eliminated the need for per-target fuzzer selection by building a runtime coordinator that measures each fuzzer's coverage.
- Implemented the orchestration layer in Python using Linux cgroups for CPU pinning; integrated 12 fuzzers (AFL, AFLFast, MOpt, RedQueen, QSYM, Angora, and others) under a unified coverage-feedback interface.

[4] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves** | Paper | Talk | Demo **USENIX 2017**

- Demonstrated the first ROP attack against Intel SGX: exploited the deterministic AEX fault oracle to probe enclave memory word-by-word and reconstruct a gadget map *without any binary knowledge* and showcased full key exfiltration.
- Built the AEX-based memory scanner and ROP-chain generator in C/x86 assembly; reverse-engineered Intel SGX library (e.g., EPCM/TCS/SSA layouts) to locate gadgets and craft the exploit entirely from outside the enclave.

[5] **SGX-Bomb: Locking Down the Processor via Rowhammer Attack** | Paper | Hacker News | Demo **SysTEX 2017**

- Showed that an enclave can weaponize Rowhammer to permanently lock the Memory Encryption Engine (MEE) and force a machine reboot — rendering SGX's integrity guarantees a denial-of-service vector for cloud multi-tenant scenarios.
- Wrote a kernel driver to reverse-engineer DRAM bank/row mapping from physical addresses; implemented the double-sided Rowhammer loop in x86 assembly inside an enclave to flip MEE metadata bits and trigger a machine lockdown.

[6] **PrivateZone: Providing a Private Execution Environment using ARM TrustZone** | Paper **TDSC 2016**

- Extended ARM TrustZone to protect arbitrary user-space processes without porting them to the Secure World, by intercepting Linux scheduling events in Monitor mode and switching per-process stage-2 page tables on each context switch.
- Implemented the stage-2 page table allocator and the Monitor-mode context-switch handler in ARM assembly; modified the Linux scheduler to emit SVC calls that trigger the switch between trusted and untrusted page tables.

[7] **OpenSGX: An Open Platform for SGX Research** | Paper | Wikipedia: SGX **NDSS 2015**
- Lowered the barrier to SGX research by building a QEMU instruction-level emulator that faithfully reproduced EPCM, SECS, TCS, and SSA data structures, enabling enclave development without SGX-enabled hardware.
- Implemented `EENTER`/`EEXIT` context-switch stubs in x86 assembly, EPCM management in QEMU's softmmu, and a `sgxlib` user library with a stub-call layer that routes enclave system calls through an untrusted trampoline.

[8] **PRIME+RETOUCH: When Cache is Locked and Leaked** | Paper **arXiv**
- Identified a blind spot in eviction-based defenses: Tree-PLRU metadata can be manipulated to "retouch" a cache set so the victim line survives eviction silently, bypassing Intel TSX-based monitors and Apple M1 PMU counters.
- Reverse-engineered the Tree-PLRU state machine on recent Intel and Apple M1 caches; built the PRIME+RETOUCH PoC in x86/ARM assembly using pointer-chasing and memory barriers; demonstrated cross-core and cross-process secret recovery.

[9] **Survey On DeFi Users' Perception of Security Breaches and Countermeasures** | Paper **USENIX 2024**
- Revealed that DeFi victims overwhelmingly skip post-incident security review and instead seek new platforms to recover losses quickly, suggesting industry regulation is more effective than user education alone.
- Conducted semi-structured interviews and surveys; performed qualitative coding of transcripts using Dedoose.

[10] **A Rule Extraction Method Using Relevance Factor for FMM Neural Networks** | Paper **KTSDE 2013**
- Enhanced the Fuzzy Min-Max (FMM) neural network by incorporating the frequency of features in training.
- Reduced classification ambiguity by weighting feature relevance and frequency during training.

[11] **Apparatus and method for open and private IoT gateway using Intel SGX** | Patent **KR102162018B1**
- Designed a secure IoT gateway utilizing Intel SGX for protocol conversion, encrypted data transfer, and enclave-based key management across cross-platform communication channels.

SOFTWARE ARTIFACTS

Autofz:	https://github.com/sslab-gatech/autofz	⑂12	★85
OpenSGX:	https://github.com/sslab-gatech/opensgx	⑂82	★308
SGXBomb:	https://github.com/sslab-gatech/sgx-bomb	⑂4	★18

TECHNICAL SKILLS

Software: C, C++, Rust, Scala, Python, TF-A, TF-RMM, Caliptra FW, PSC FW (NVIDIA IROT), Linux Kernel, KVM, QEMU, ARM/X86 Assembly, Fuzzing, GDB, CoCLI

Hardware: GEM5 (x86 simulator), Processor Pipeline, Cache, TLB, Microcode, HW Side Channel, (System)Verilog, Chisel (RISC-V), Intel SGX & TDX, ARM CCA & TrustZone, IOMMU, PCIe

Spec: Entity Attestation Token (EAT), Concise Reference Integrity Manifest (CoRIM), DICE Protection Environment (DPE), Security Protocol and Data Model (SPDM), TEE Device Interface Security Protocol (TDISP)

PROFESSIONAL ACTIVITIES

[1] **Journal Review**
- IEEE Transactions on Dependable and Secure Computing — Impact Factor: 7.3, *2024*
- Computer & Security — Impact Factor: 5.7, *2024*
- Journal of Information Security and Applications — Impact Factor: 5.6, *2024*
- IEEE Access — Impact Factor: 3.9, *2024*
- ACM Transactions on Internet Technology — Impact Factor: 3.9, *2024*
- IEEE Journal of Communications and Networks — Impact Factor: 3.6, *2024*

[2] **Conference Review**
- IEEE Mobile Ad Hoc and Smart Systems (MASS) — *2024*
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) — *2024*

[3] **External Reviewer**
- IEEE Symposium on Security and Privacy (Oakland) — *2016, 2017, 2018, 2020, 2021, 2022*
- USENIX Security Symposium (Security) — *2015, 2021, 2022, 2023, 2024*
- ACM Conference on Computer and Communications Security (CCS) — *2015, 2017, 2018, 2019, 2023*
- Network and Distributed System Security Symposium (NDSS) — *2019, 2020, 2021*
- ACM ASIA Conference on Computer and Communications Security (ASIACCS) — *2015, 2016, 2018*
- ACM Symposium on Operating Systems Principles (SOSP) — *2021*
- ACM International World Wide Web Conference (WWW) — *2018*

INVITED TALKS

[1] **A Novel Cache Side Channel Attack without Attacker Eviction**
- Presented at Intel Side Channel Academic Program (SCAP), Virtual, Sep 2020

[2] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves**
- Presented at the 26th USENIX Security | Presentation Link, Vancouver, BC, Canada, Aug 2017
- Presented at KimchiCon | Presentation Link, Daejeon, South Korea, Aug 2017

[3] **Tutorial: Trusted Execution Environment: TrustZone, ITP and SGX**
- Korea Institute of Information Security and Cryptology, Seoul, South Korea, Apr 2015

EDUCATION

Institution	Degree	Field of Study	Dates	GPA
Korea Advanced Institute of Science and Technology	*Ph.D.*	Information Security	Aug 2015 – Feb 2021	3.95
Korea Advanced Institute of Science and Technology	*M.S.*	Information Security	Sep 2013 – Aug 2015	4.1
Handong Global University	*B.S.*	Computer Science	Mar 2010 – Aug 2013	4.2

THESIS

Unintended Consequences of System Design: Unpremeditated Usage of Benign System Components Devastates Security	Ph.D. Thesis.
Detecting Emulated Environment: Exploiting Behavioral Discrepancies in QEMU	M.S. Thesis.